

Vectors: Components and Operations



Vector arithmetic

- 1 Let $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 4 \rangle$. Find:

a $\mathbf{u} + \mathbf{v}$

b $3\mathbf{u} - 2\mathbf{v}$

c $|\mathbf{u}|$

- 2 Find the component form and magnitude of the vector from $P(1, 3)$ to $Q(5, -2)$.
- 3 Find the unit vector in the direction of $\mathbf{v} = \langle 6, -8 \rangle$.
- 4 Write the vector of magnitude 12 making an angle of 150° with the positive x -axis in component form (exact values).

Applications

- 5 A plane flies due east at 200 mph while the wind blows due north at 50 mph.
- a** Find the magnitude of the plane's resultant velocity, to one decimal place.
- b** Find the direction of the resultant, as an angle north of east to two decimal places.
- 6 Two forces $\mathbf{F}_1 = \langle 30, 0 \rangle$ N and $\mathbf{F}_2 = \langle 0, 40 \rangle$ N act on an object. Find the magnitude of the resultant force.
- 7 A boat aims to cross a river by heading due north at 4 m/s, but the current pushes it east at 1.5 m/s.
- a** Find the boat's resultant speed, to two decimal places.
- b** If the river is 200 m wide, find how long the crossing takes.
- c** Find how far downstream (east) the boat lands.

Position vectors and geometry

- 8 Given $A(-2, 5)$ and $B(4, -3)$, find the midpoint of $\overline{AB}$ and the vector $\vec{AB}$.
- 9 Let $\mathbf{a} = \langle 2, 3 \rangle$ and $\mathbf{b} = \langle -4, 1 \rangle$. Find a vector $\mathbf{c}$ such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \langle 0, 0 \rangle$, and verify your answer.
- 10 Express $\mathbf{v} = \langle 5, -12 \rangle$ in terms of the standard unit vectors $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{v}|$.

More applications

- 11 An airplane has an airspeed of 305 mph and flies on a heading 7.5° west of due north, while a 40 mph wind blows due east. Find the plane's resultant ground velocity vector, and its ground speed, to one decimal place (take east as the positive x -direction and north as the positive y -direction).
- 12 Given $\mathbf{u} = \langle 4, -1 \rangle$, find a vector orthogonal to $\mathbf{u}$ with the same magnitude as $\mathbf{u}$.